Customer Segmentation Using Multi-Paradigm Clustering

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Abstract

This study uses methods that examine data without initial categories to segment customer data for a retail enterprise. The approach identifies groups showing different patterns in consumption and demographic features. The analysis examines data features and develops these features through processing. The study then applies analysis that reduces dimensions to address complexity in data with multiple variables. The analysis compares different methods including one approach using means, a second approach that builds groups through steps, and a third method using relationships between data points. This comparison and assessment leads to a final structure that contains four distinct clusters. The results show four main customer groups that differ in characteristics: customers with high value and elite status, customers at a middle level who purchase wine products actively, customers who are sensitive to budget and are in extended families, and customers with low income who are independent.

1 Introduction

In the retail environment that exists at the current time, the main strategic focus for enterprises shows a shift from marketing approaches that target broad populations toward approaches that provide high levels of individual targeting. The concept of customer personality traits is important for businesses that aim to improve product offerings and increase efficiency in marketing. Companies identify specific needs that different consumer groups have and examine shopping patterns and demographic details that these groups show. This identification allows companies to move from advertising campaigns that are costly and have broad reach toward marketing strategies with specific targets. These strategies reach audiences that show high likelihood for purchase with more effectiveness.

The main objective that this study follows is to achieve customer segmentation at a detailed level. The study applies methods using machine learning without supervision to a retail dataset that is comprehensive. The data that the analysis uses come from the Customer Personality Analysis that Patel [1] proposed in 2021. This analysis includes four dimensions. The first dimension examines demographic characteristics that include Income, Education, and Family Size. The second dimension examines expenditure across specific categories that include Wines, Meat, and Gold. The third dimension examines responsiveness to promotional campaigns through involvement in five marketing activities. The fourth dimension examines preferences for purchasing channels that include online, in-store, and catalogue options.

This work uses the approach that Kapoor [2] presents to develop a method for the analysis. The method applies an approach that reduces features to show main patterns in the data. The approach reduces the initial thirty-two features to three components that contain the main patterns, and this addresses the issue that high numbers of features present. Following this, the work compares different approaches for identifying groups, including methods that use means, methods that combine cases in stages, and methods that examine patterns in relationships. The comparison indicates that four separate groups provide a stable way to describe the different types in the data. The four groups show different features that distinguish them. This work examines the features that each group shows to provide a basis for decisions that organizations can use. The analysis suggests that organizations can apply the findings to increase the extent that customers remain engaged and to improve the outcomes that marketing efforts produce relative to the resources that these efforts require.

2 Related Work

Customer segmentation represents work that research in markets has used for considerable time, and approaches that are traditional rely in large part on analysis using recency, frequency, and monetary factors. Data from large sources and the development of methods for learning without supervision have produced change in the field that moves toward models of behavior in consumers that show increasing levels of complexity.

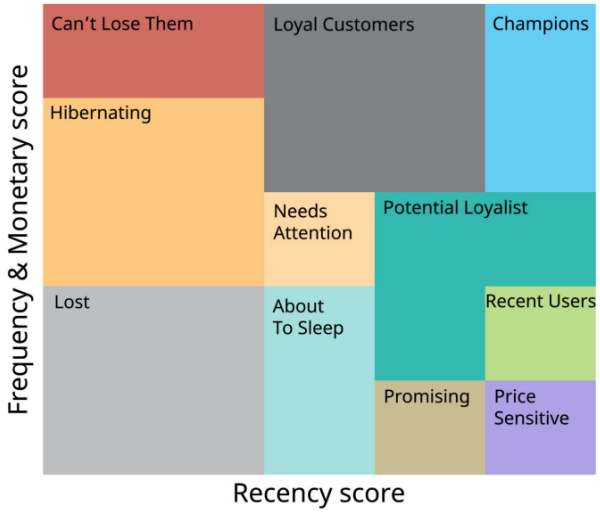


Figure 1: Framework of RFM Scores (Source: Vecteezy [3])

Traditional Clustering Approaches. The method for forming groups using means continues to provide the main approach for efforts in segmentation of consumers, and this approach receives value mainly from efficiency in processing data and from features that allow clear interpretation. Use of this method continues in

wide form but the method also shows limitations that are notable, and these include high levels of response to data points that are anomalous and also include assumptions about groups that are convex and spherical in form [4].

Dimensionality Reduction in Marketing. Data from retail sources often contain high numbers of dimensions, and these data face the problem termed the curse of dimensionality, which is a condition in which measures of distance that are conventional show reduced power for making distinctions. This approach not only reduces the number of dimensions but also reduces problems from relationships between features that show high correlation, and these effects improve clarity in the signal that the data contain.

Graph-based and Hierarchical Extensions. Work in recent scholarship has moved beyond limitations in methods using centroids such as the means method, and this work has examined approaches for forming groups using spectral features and approaches using agglomeration. The approach using spectral features operates on values that are eigen values from a matrix termed Laplacian, and this approach shows high capability for identifying segments that data connectivity and structures in the underlying manifold define [5]. The approach using hierarchies in forming groups provides a different type of value, which is that this approach organizes the base of customers in a form that is nested and similar to a tree structure. This form of presentation allows individuals conducting analysis to follow how segments that are finer in form combine to produce categories of consumers that are broader, and this capability provides a perspective of the market that operates at multiple levels of resolution.

3 Methodology

This study develops a framework for customer segmentation that does not use labels. The approach combines methods for developing features, reducing dimensions, and using multiple clustering methods. It provides a balance between statistical rigor and practical application.

3.1 Feature Engineering and Preprocessing

The work transforms raw transaction data into behavioral features through a structured preprocessing approach that ensures statistical reliability.

3.1.1 IQR-based Outlier Filtration. The **Inter quartile Range (IQR)**, uses the difference between the 75th (Q_3) and 25th (Q_1) percentiles, which is a measure resistant to extreme values[6]. The filtration process removes records outside the range from the $[Q_1 - 1.5 \times \text{IQR}, Q_3 + 1.5 \times \text{IQR}]$.

3.1.2 Robust Scaling. Financial features often follow distributions with heavy tails. Standard approaches to normalization are inadequate for these distributions. The transformation **Robust Scaling** divides the difference relies on the median and the IQR:

$$x_{scaled} = \frac{x_i - \text{median}(x)}{Q_3 - Q_1} \tag{1}$$

3.1.3 Z-score Normalization. Following the clustering process, **Z-score Normalization** using standard deviations aids in comparing segments. This method rescales features to standard deviations from the overall mean. ($\bar{x}_j$):

$$Z_{k,j} = \frac{\mu_{k,j} - \bar{x}_j}{\sigma_j} \tag{2}$$

Here, $\mu_{k,j}$ indicates the mean of feature j in cluster k.

3.2 Dimensionality Reduction: PCA

To address issues relating to high variable counts and relationships between measures in data, the approach uses a method that changes data into a new structure. Principal Component Analysis(PCA) provides an orthogonal linear form that presents data in reduced form [7]. The method works by finding components in sequence. Each component that follows the previous component shows the maximum variance that remains in data. The study retains the first three components from this analysis. This approach increases information contained in the reduced form and allows examination in three dimensions.

3.3 Clustering Paradigms

The analysis uses three different approaches for finding groups. This strategy that applies multiple methods provides confidence that groups show stability and clear separation between forms.

3.3.1 K-Means Clustering. K-Means works to divide observations into groups by reducing the measure that shows within-group differences. The measure use Within-Cluster Sum of Squares(WCSS):

$$J = \sum_{j=1}^{k} \sum_{i=1}^{n} \|x_i^{(j)} - \mu_j\|^2 \tag{3}$$

μ_j indicates the center point for that group. This approach shows high speed in analysis and works well for finding spherical group forms in data following the component analysis.

3.3.2 Agglomerative Hierarchical Clustering. The second method uses a hierarchical approach that combines observations in stages. Ward Linkage provides the basis for this analysis [8]. The method focuses on reducing total within-group variance when combining groups at each stage. This strategy that works from individual observations upward develops a hierarchical structure that shows the full sample. This view reveals how smaller groups that contain specific features combine to form larger categories that show more general patterns.

3.3.3 Spectral Clustering. The third method addresses possible nonlinear structures in data. Spectral Clustering provides value for finding these forms. The approach develops a k-nearest neighbors affinity matrix A and examines the **Graph Laplacian** $L = D - A$, This method finds groups using data connectivity rather than simple distance measures in space.

3.4 Model Selection and Validation Metrics

The analysis requires determining the optimal number of groups K and assessing the quality of divisions that result. This assessment uses a framework that examines multiple criteria.

3.4.1 Elbow Method. The approach **Elbow Method** that examines the measure of WCSS considers different values for the number of groups. The point that shows a clear change in the rate of decrease

indicates the number that provides the main benefit, and additional groups beyond this point show limited improvement in the measure of variance that the analysis explains.

3.4.2 Eigenvalue Gap Analysis. Analysis of the gap between values provides further support for selecting the number of groups in approaches using graph structure. The theory that Von Luxburg [9] presents that the optimal K aligns with the index that maximizes the gap between consecutive eigenvalues λ_K and λ_{K+1} of the normalized Laplacian matrix.

3.4.3 Silhouette Coefficient. The **Silhouette Coefficient (s)** that Rousseeuw [10] describes provides assessment of the overall quality of the grouping that the analysis produces:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \tag{4}$$

$a(i)$ and $b(i)$ denote intra-cluster cohesion and nearest-cluster separation for point i, respectively. A mean value across all points that approaches one indicates that the analysis produces groups with clear separation and internal consistency.

4 Experiments

4.1 Feature Engineering and Data Preparation

This approach was designed to reveal the patterns relating to spending that appear in the main attributes. At the stage for the main processing, field names received changes that provided more clear forms. Identifiers that were difficult to interpret, such as *Mnt* and *Num...Purchases*, were changed to forms that were more direct, such as *Wines* and *Web*. The study used the final date in the data (December 2014) as the reference point, and this allowed establishing 2015 as the year for developing the measure of customer *Age*. Attributes relating to demographic features also received consolidation in this phase. Five different levels that indicated education were combined to form three main categories: *Undergraduate*, *Graduate*, and *Postgraduate*. Various forms that indicated marital status were simplified to provide two clear categories. These categories were *Alone* (this includes single, divorced) and *Couple* (married, together). This consolidation reduced the dimensions that were present in the categorical data.

The phase for developing the main features involved creating indicators that showed more complex patterns in behavior. This was achieved by combining multiple measures. Expenditures that appeared across six main categories were aggregated in the analysis. These categories included wines, meats, gold products, and others. The aggregation produced the measure of total expenditure (*Spent*). The study also combined counts that indicated children and teenagers to form a single variable (*Children*). This variable then provided the basis for developing the measure of total *Family_Size* and for creating a measure that indicated the presence of parents (*Is Parent*). The analysis also examined sensitivity to marketing approaches. The total that showed accepted offers across five campaigns was developed to form the measure of *Total Accepted Cmp*.

The study introduced several measures to show more specific patterns in consumption:

- **RFM-Inspired Measures:** The analysis aggregated purchases that occurred through web, catalog, and store channels. This aggregation produced the measure of total textit-Frequency. The measure then allowed developing the form that indicated Average Order Value (*Avg Order Value*).

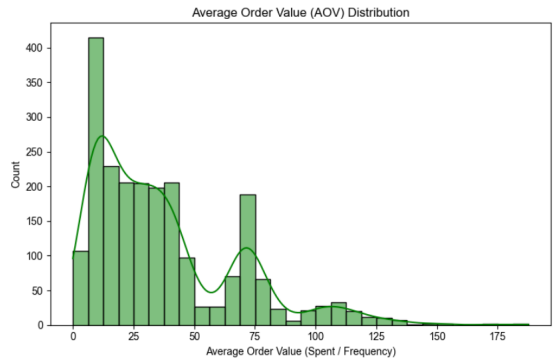


Figure 2: Distribution of Average Order Value (AOV)

- **Discounts Preferences and Channel Allocation:** Measures the ratio of purchases using discounts (*Deals_Ratio*) and separate ratios for catalog, store, and web channels. These measures provide more specific information about the pathways customers use for shopping.
- **Product Preferences:** For the six main product categories, the proportion of total spending in each category (the *Pcnt* series) was calculated. This approach using proportions removes the effect of differences in absolute income levels and allows the model to focus on actual preferences between categories.

For **outliers handling**, after entries with missing data were removed, a specific filtering approach was applied. This approach removed extreme cases where *Age* exceeded 114 years, set a limit for *Income* at or below $\leq 600,000$, and removed records that were logically inconsistent. Examples of inconsistent records include those with a ratio of spending to income exceeding 0.7 or discount ratios that appear implausible. This comprehensive approach to preparing data, combined with the removal of redundant features such as *ID* and *Z_CostContact*, produced a feature space with high information density and low noise.

4.2 Data Pre-processing Pipeline

For the pipeline to process data before analysis, using a *ColumnTransformer* to address the heterogeneous structure of the dataset. Features were directed into three separate processing streams:

- **Numerical Variance Normalization:** For continuous variables such as *Income*, *Spent*, and the *Pcnt* series, the *RobustScaler* was applied to normalize numerical variance. This method differs from conventional normalization using Z-scores because it relies on the range between quartiles (IQR) for scaling. This approach enhances robustness of the model against outliers that occur infrequently and extreme figures for income that remain in data.

- **Categorical Encoding:** For qualitative attributes such as *Education* and *Marital_Status*, categorical encoding was performed. These attributes were converted into a format that is machine-readable using *One-Hot Encoding*. This transformation captures the effect of demographic status on consumption behavior that is non-linear.
- **Binary Pass-Through:** Binary indicators, including *Is_Parent* and *Complain*, and indicators that show response to *Complain* from before, including *Accepted Cmp1-5*, remain in the same form. This approach that uses the same form allows these indicators to maintain clear meaning in the space with multiple dimensions that results from the process.

4.3 Dimensionality Reduction and PCA

The features that result from this process, which include 32 variables after the process that converts categories to numerical form, require reducing the number of dimensions. **Principal Component Analysis (PCA)** provides this reduction. This method reduces the data to show the main patterns in forms that do not relate to each other.

- **Variance Analysis and Scree Plot:** Results from this analysis indicate that the first three components ($N = 3$), with N equal to three, explain **53.37%** of of the total variance (PC1: 29.07%, PC2: 15.89%, PC3: 8.41%). Variance shows distribution across multiple dimensions, but this reduction to three dimensions was selected to provide efficiency while maintaining the main structures in the population.
- **Feature Weight Interpretation:** This analysis reveals dimensions of behavior that were not directly measured:
  - **PC1 (Economic Capacity):** This component shows positive weights from *Avg_Order_Value* (+0.393) and *Spent* (+0.308), and negative weights from *Family_Size* (-0.323) and *Deals _Ratio* (-0.291), This pattern indicates a relationship between spending among individuals with higher income and limitations on budgets that appear in households that are larger.
  - **PC2 (Product Preference):** This component shows a clear difference between *Fruits/Fish* (+0.467, +0.412) and *Wines* (-0.408), This dimension separates preferences that relate to different approaches to living.
  - **PC3 (Channel Preference):** shows strong opposition between *Web_Ratio* (+0.665) and *Store_Ratio* (-0.567), This component identifies the continuum between digital shopping patterns and physical shopping patterns in the study population.

4.4 Clustering Analysis

The clustering pattern analysis is based on the three-dimensional feature space expansion after PCA dimensionality reduction. This study adopts a multi-strategy parallel analysis method, aiming to reveal potential consumer groups. This stage covers the determination of the number of clusters, the comparison and execution of multiple algorithms, and the verification of spatial validity.

Optimal Cluster Cardinality (K). The value of the optimal number of clusters $K$was verified through three different mathematical indicators, all pointing to $K = 4$as the best choice. Firstly,

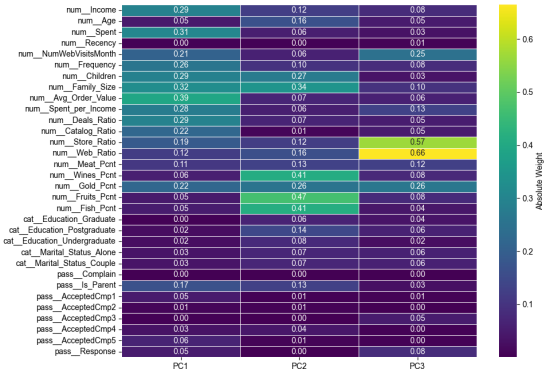


Figure 3: Feature Weight Heatmap for the 3 Principal Components

the *Elbow Method* shows that the gradient of the sum of squares within the cluster (WCSS) has a significant turn at $K = 4$. Secondly, the *Silhouette Analysis* conducted on the value of Kranging from 2 to 10 shows that there are obvious peaks when there are 4 clusters, indicating that the cohesion and separation degree of the data are most balanced at this time. Finally, the hierarchical clustering tree diagram (Dendrogram) using the Ward linkage method (figure 4) presents 4 clear main branches at a distance of approximately 40 points from the threshold. These indicators jointly support the decision to select 4 clusters.

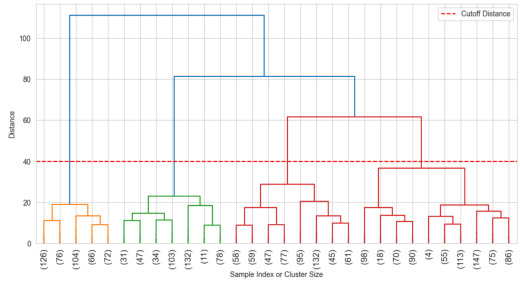
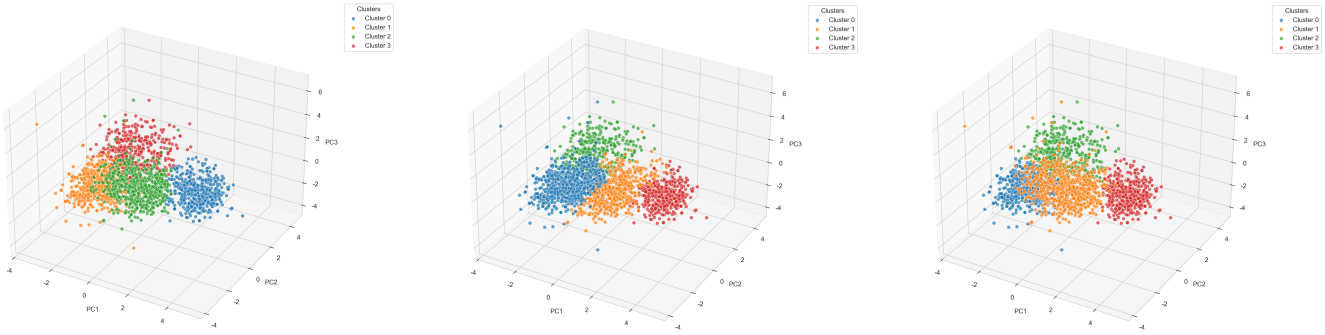


Figure 4: Hierarchical Clustering Dendrogram

Algorithmic Implementation and Performance. To ensure the robustness of the segmentation results, this study ran three clustering algorithms on the data after PCA dimensionality reduction. The first one is Agglomerative Clustering, which initially reveals the hierarchical relationship among samples by minimizing the differences within clusters. The second type is K-Means clustering. Its contour coefficient shows the highest among the three methods, reaching 0.3246, indicating that the data after dimensionality reduction forms clusters with clear structures and mutual independence. The third type is Spectral Clustering, which mines potential patterns through the similarity connections among samples. Eigenvalue analysis shows that there is a significant jump at the fourth eigenvalue, further confirming the rationality of the four-cluster division. Although the contour coefficient of this method is slightly lower (0.3185), its clustering results show extremely strong stability, and the scale distribution of each cluster is highly consistent with that of other algorithms.

Spatial Validation and Convergence. The study provides confirmation through visual examination. Results from the grouping appear on the first three components from the analysis. Figure 5 shows three-dimensional plots that reveal consistent divisions in space across all three approaches. This occurs despite different underlying methods. The approach using centers shows the most clear circular patterns with high density. The approach using connections and the approach using structure support these divisions from different perspectives. The first examines patterns in connections between points. The second examines differences within groups.



(a) K-Means Clustering (b) Hierarchical Clustering (c) Spectral Clustering

Figure 5: 3D Visualization of Clustering Results

4.5 Comparative Analysis and Model Robustness

To establish that groups identified in data reflect patterns in the data rather than results from the approach, comparison across approaches was conducted. The comparison that Table 1 presents indicates mean values for main factors—*Income*, *Spent*, and *Wines_Pcnt*—show consistency across the three approaches. Consistency matrices provide further support for this robustness. The Wine Lovers group showed alignment between K-Means and Spectral approaches with a sample of six hundred thirty-four. K-Means was selected as the model for profiling individuals in the study. This approach shows stronger internal consistency and clearer features for interpretation.

Table 1: Cross-Algorithmic Feature Comparison and Stability Analysis ($K = 4$)

Feature (Mean)	K-Means	Hierarchical	Spectral	Overall Mean
High-Value Elite	(Cluster 0)	(Cluster 3)	(Cluster 3)	–
Population %	23.1%	20.1%	20.9%	22.0%
Income ($)	76,784	78,168	77,706	52,001
Spent ($)	1,389.3	1,433.3	1,424.3	606.9
Wine Lovers	(Cluster 2)	(Cluster 1)	(Cluster 1)	–
Population %	28.8%	26.0%	34.3%	29.7%
Wines %	0.61	0.56	0.60	0.46
Web Ratio	0.38	0.35	0.36	0.33
Budget Families	(Cluster 1)	(Cluster 0)	(Cluster 0)	–
Population %	26.2%	34.2%	23.1%	27.8%
Deals Ratio	0.39	0.36	0.39	0.24
Family Size	3.46	3.28	3.50	2.59

4.6 Multi-dimensional Customer Segmentation

Using features relating to population characteristics, patterns of use, and responses from the K-Means analysis, the study identifies four separate groups of individuals. Figure 6 presents a chart using measures that are adjusted to show differences in preferences across six main product categories.

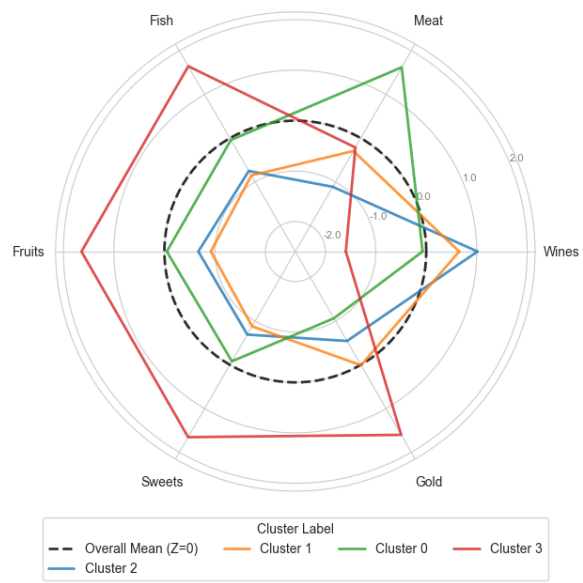


Figure 6: Product Preference Radar chart by Cluster

4.6.1 High-Value Elite Group (Cluster 0). This group represents individuals with the highest income levels. The group shows average income of $76,784 and a low rate of individuals with children at 7.5%.

Use by this group focuses on high-quality protein products. Meat purchases by the group reach $470. This represents 34% of total spending by individuals. The group shows high levels of response and represents individuals described as Quality individuals in the analysis. These individuals show strong preference for **Catalog** methods of purchase at a rate of 31%. The group also shows active response to promotional efforts with acceptance at 73%. Sensitivity to price appears low in this group. Evidence for this is the minimal ratio for *Deals_Ratio* at 0.06.

4.6.2 Active Wine Lovers Group (Cluster 2). This group contains established households, with parents at 94%, and shows solid financial standing at $59,206 in income. The group directs a substantial share of spending toward **Wine**, with 61% of total expenditure in this category. These individuals show patterns that indicate a digital-leaning approach using balanced methods, with a clear preference for **Web** purchases that show a ratio of 38%. Response to promotions appears moderate at 12.6%. This segment demonstrates sophisticated preferences that combine with the convenience that online shopping provides.

4.6.3 Budget-Sensitive Large Families (Cluster 1). The Budget Sensitive Large Families in Cluster 1 show patterns that indicate these individuals function as Deal Hunters. This cluster shows the largest households, with a mean size of 3.46, and the highest ratio of dependency at 1.70 children. The group shows income below average at $40,497 despite educational attainment that appears relatively high. Consumption patterns appear utilitarian and show price-driven features, with the highest ratio for *Deals_Ratio* at 0.39. These individuals rely heavily on physical **Store** visits, showing a ratio of 59% to use advantages from discounts. Engagement with premium campaigns appears limited.

4.6.4 Low-Income Independent Spenders (Cluster 3). The Low Income Independent Spenders in Cluster 3 function as Gold Seekers. This segment shows the youngest mean age at 39.3 and shows the entry-level market position. The group shows the lowest average income at $30,143. The segment allocates a substantial share, at 22%, of the budget to **Gold** products while largely avoiding wine. These individuals show the highest brand friction, with the highest rate of complaints at 0.012 and minimal interaction with campaigns. This likely indicates that specialized efforts for building trust and incentives at entry-level appear required.

4.6.5 Summary of Consumer Segmentation. The convergence of results from multiple clustering algorithms confirms four robust profiles: (1) **High-Value Elite** (Child-free, quality-driven, catalog-loyal); (2) **Active Wine Lovers** (Parental, web-centric, connoisseurs); (3) **Budget-Sensitive Families** (Large households, store-loyal, deal-seekers); and (4) **Independent Spenders** (Younger, gold-focused, high-friction).

5 Conclusion

The research uses process that reduces elements in data and applies multiple forms of analysis that find patterns. Using a scaling procedure that provides resistance to extreme values and a procedure that removes values outside normal range was important for data that includes financial measures. The analysis produced three main factors that show relationships between capacity to purchase, focus on particular products, and preference for different means of purchase. Results from multiple procedures that find groups in data indicate similar patterns.

The analysis reveals four separate groups. The first group includes individuals with high value who focus on quality and use catalog means for purchase. The second group includes individuals who purchase wine frequently and use digital means. The third group includes large families that show sensitivity to price and prefer physical stores and use discounts. The fourth group includes younger individuals with lower income who show interest in gold products and require more service. These findings allow strategies that provide specific approaches for different groups.

The approach that this work presents provides a method that organizations selling products can use for understanding features of individuals. Work that follows this study could use data that follows individuals over time. This would show movement between groups. The work could also use models that provide predictions for the value that individuals provide over time. These developments would allow organizations to maintain results that support continued success in markets that show increasing division and competition.

AI Usage Statement

During work on this report, AI tools provided assistance. I used Gemini for support with translation, Latex formatting suggestions and initial collection of sources. These tools improved clarity in text and correctness in form. The main concepts, analysis of data, central claims, and interpretation of findings represent work that the author completed. The author accepts responsibility for accuracy and quality of content that this report presents.

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